Business Requirement Document (BRD)

Project: Customer Churn Analysis & Retention Strategy
Domain: Banking Analytics
Version: 1.0
Author: Nasrul Khair (Data Analyst)

1. Project Overview

This project aims to analyze bank customer data to identify the key factors influencing customer churn and provide data-driven retention strategies. The analysis will support management in understanding customer behavior, churn patterns, and opportunities for improving engagement.

Dataset Summary

- Demographics (Age, Gender, Geography)
- Financial Indicators (Balance, Credit Score, Estimated Salary)
- Banking Behaviors (Tenure, Activity Status, Credit Card Ownership)
- Customer Status (Exited or Retained)

Attribute	Detail
Source	Local Folder (Bank_Churn.csv)
Rows	10,001
Type	Structured Tabular Data
Destination (Raw layer)	Snowflake Data Warehouse – Schema: RAW.BANK_CHURN

2. Business Objectives

1. Identify characteristics of customers likely to churn.
2. Quantify churn rate and segment customers by risk level.
3. Develop actionable insights for retention strategy.
4. Enable management to monitor churn KPIs through Power BI dashboards.
5. Establish a repeatable data pipeline from ingestion to reporting.

3. Business Questions

(To be refined during Sprint 2 – EDA)

1. What is the overall churn rate across all customers?
2. How do age, tenure, and balance influence churn likelihood?
3. Are inactive customers or those without credit cards more likely to churn?
4. Which geography or gender shows higher churn trends?
5. What common patterns exist among high-balance but low-activity customers?
6. How do credit score segments correlate with churn?
7. What is the retention trend by tenure segment?

4. Data Gathering

Please use the following data related to Bank Customer and Associated details:

- Source Name: Bank_Churn.csv
- Description: Customer churn dataset including demographics, credit, and account info
- location: Local folder
- NumOfRows: 10,001

Column Name	Description	Example	Data Type	Business Use
RowNumber	Record Index	1	Integer	Ignored in analysis
CustomerId	Unique customer ID	15634602	Integer	Identifier for customer
CreditScore	Credit rating	619	Numeric	Measure of customer creditworthiness
GeographyId	Location (1=France, 2=Spain, 3=Germany)	1	Categorical	Region segmentation
GenderId	Gender (1=Male, 2=Female)	2	Categorical	Gender-based insights
Age	Customer age	42	Numeric	Key predictor for churn
Tenure	Years with bank	7	Numeric	Measures customer loyalty
Balance	Account balance	0	Numeric	Financial indicator
NumOfProducts	Products owned	1	Numeric	Measures engagement
HasCrCard	Has credit card (1=Yes, 0=No)	1	Binary	Product ownership
IsActiveMember	Active member (1=Yes, 0=No)	1	Binary	Engagement level
EstimatedSalary	Estimated annual income	101348.88	Numeric	Income category
Exited	Whether customer left (1=Yes, 0=No)	1	Binary	Churn indicator
Bank DOJ	Date joined bank	23/3/2016	Date	Used for tenure validation

5. Data Dictionary

- RowNumber – correspond to the record number and has no effect on the output
- CustomerId – Contains random values and has no effect on customer leaving the bank
- Surname – the surname of a customer has no impact on their decision to leave the bank
- CreditScore – can have effect on customer churn, since a customer with a higher credit score is less likely to leave the bank.

Credit Score

- Excellent: 800 – 850
- Very Good: 740 – 799
- Good: 670 – 739
- Fair: 580 – 669
- Poor: 300 – 579

- Geography – a customer's location can affect their decision to leave the bank.
- Gender – it's interesting to explore whether gender plays a role in a customer leaving the bank.
- Age – age certainly relevant, since older customers are less likely to leave their bank than younger ones.
- Tenure – refers to the number of years that the customer has been a client of the bank. Normally older clients are more loyal and less likely to leave the bank.
    - Balance – also a very good indicator of customer churn, as people with a higher balance in their accounts are less likely to leave the bank compared to those with lower balances.
    - NumOfProducts – refer to the number of products that a customer has purchased through the bank.
    - HasCrCard – denotes whether or nota customer has a credit card. This column is also relevant, since people with a credit card are less likely to leave the bank.
        - 1 represents credit card holder
        - 0 represents non credit card holder

    - IsActiveMember – active customers are less likely to leave the bank
        - 1 represents Active Member
        - 0 represents Inactive Member
    - Estimated Salary – as with balance, people with lower salaries are more likely to leave the bank compared to those with higher salaries
    - Exited – whether the customer left the bank
        - 1 represents Exit
        - 0 represents Retain
- Bank DOJ – date when the customer joined the bank

6. Business Rules

Rules	Description
Churn Definition	Customers with Exited = 1 are churned.
Activity	IsActiveMember = 0 → Low engagement risk group.
Location Mapping	1=France, 2=Spain, 3=Germany
Gender Mapping	1=Male, 2=Female
Data Validation	Exclude rows with missing or invalid CustomerId.
Date Handling	Convert Bank DOJ to datetime and validate Tenure.

- Customers with Exited = 1 are considered churned.
- IsActiveMember = 0 may indicate low engagement.
- GeographyID maps to: 1=France, 2=Spain, 3=Germany.
- Exclude customers with missing or invalid CustomerId.
- Convert Bank DOJ to datetime for tenure calculation validation.

7. Deliverables

Deliverables	Description	Tool/Owner	Related Epic
BRD	BRD & FRD	BA	Epic 1
Snowflake Database Setup	Create BANK_CHURN DB and schemas (RAW, CLEAN, ANALYTICS)	Data Engineer	Epic 2
Data Cleaning & Transformation Scripts	Python/SQL scripts for preprocessing	Data Engineer	Epic 2
Star Schema Data Model	Fact table: Fact_CustomerChurn + dimensions (Customer, Geography, Product)	Data Modeler	Epic 3
EDA Report	Insights and trend analysis	Data Analyst	Epic 3
Power BI Dashboard	Interactive churn dashboard with filters	Power BI Developer	Epic 4
Row-Level Security Setup	Restrict views by geography	BI Admin	Epic 5
Data Refresh Pipeline	Scheduled refresh via gateway	BI Admin	Epic 5
Documentation Package	ER diagram, data dictionary, process flow, KPIs	BA	Epic 6

8. Stakeholders & Responsibilities

Role	Responsibility	Primary Contact
Business Analyst	Gather requirements, create BRD/FRD, perform EDA	Nasrul
Data Engineer	Build Snowflake warehouse, manage data pipelines	Nasrul
Data Analyst	Conduct churn analysis, calculate KPIs	Nasrul
Data Modeller	Design analytical star schema	Nasrul
Power BI Developer	Develop dashboards, configure RLS and refresh	Nasrul
End Users	Consume dashboards, provide feedback	Nasrul

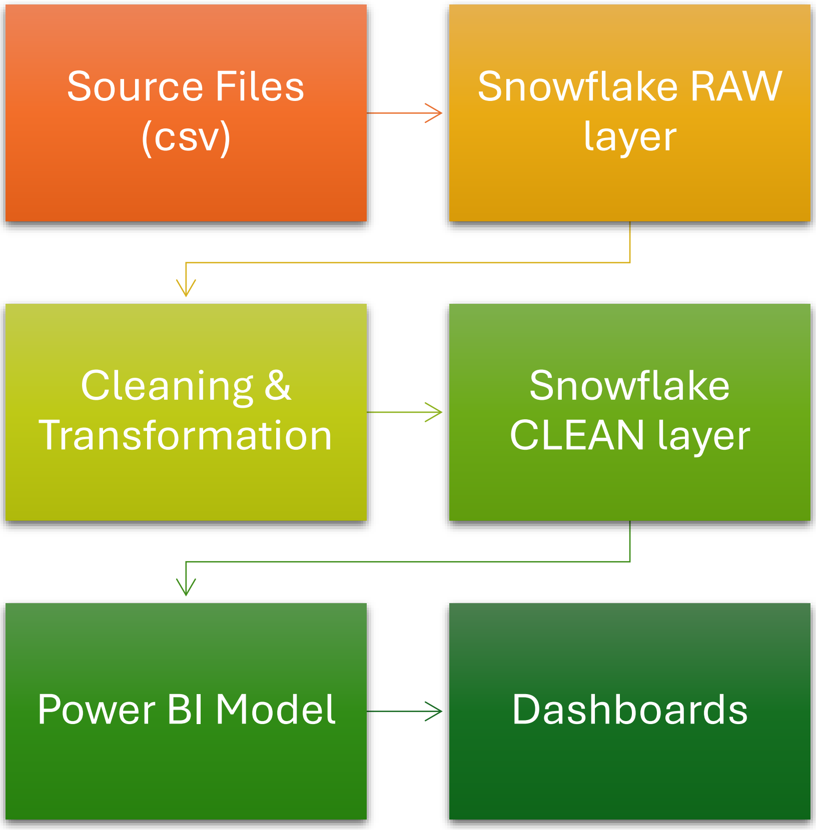
9. KPIs and Success Metrics

KPI	Description	Formula	Target / Benchmark
Customer Churn Rate	% of customers who left	*(Churned Customers / Total Customers) *100*	≤ 10%
Customer Retention Rate	% of customers retained	*100 - Churn Rate*	≥ 90%
Active Member Ratio	% of customers active in last 12 months	*(Active Members / Total Customers) *100*	≥ 70%
Avg. Credit Score (Churned vs Retained)	Compare creditworthiness	*(AVG(CreditScore_Churned) vs AVG(CreditScore_Retained)*	Retained > Churned
Avg. Balance (Churned vs Retained)	Measure financial profile difference	*AVG(Balance_Churned) vs AVG(Balance_Retained)*	Retained > Churned
High-Risk Segment %	Share of customers predicted to churn	*(Predicted_Churners / Total) * 100*	Monitor trend ↓
Dashboard Adoption Rate	% of active report users	*(Active Users / Total Users) * 100*	≥ 80%
Refresh Success Rate	% of successful scheduled refreshes	*(Success / Total Schedules) * 100*	100%

10. Dependencies & Constraints

Dependency	Description
Data Availability	Bank_Churn dataset must be uploaded to Snowflake before transformation.
Tool Access	Power BI Pro license and Snowflake connection credentials required.
Security	RLS must align with internal user access policy.
Environment	Stable gateway for automated refreshes.

11. Data Flow



12. Refresh Frequency

Source Name	Owner	Refresh Frequency
bank_churn.csv	Data Team	Weekly
Snowflake RAW Schema	Data Engineering	Daily
Snowflake CLEAN Schema	Data Engineering	Daily
Power BI Dataset	BI Team	Daily / On Refresh

13. KPI Mapping to data Fields

KPI	Calculation	Related Fields
Churn Rate	(#Exited Customers / Total Customers)	Exited
Average Credit Score	AVG(CreditScore)	CreditScore
Avg. Tenure	AVG(Tenure)	Tenure
Active vs Inactive Members	COUNT(IsActiveMember)	IsActiveMember
Product Penetration	AVG(NumOfProducts)	NumOfProducts
Avg. Balance per Geography	AVG(Balance) GROUP BY GeographyID	Balance, GeographyID

14. Sprint Deliverables

1. Business Requirement Document (BRD)
2. Listed data sources, owners, refresh frequency, and access levels
3. Created a small data flow diagram or table mapping doc

Data Engineering & Warehouse Setup

1. Warehouse & Schemas Setup

"As a data engineer, I want to set up the data warehouse to host raw and clean data layers so that data is organized and ready"

Goal: Build and configure Snowflake to host raw and cleaned banking data for analysis.

Warehouse & Schema Setup

#	Task	Description	Output
1	Create Snowflake Database and Warehouse	Create a new database (e.g., BANK_ANALYTICS_DB) and warehouse (e.g., BANK_WH)	Database + warehouse ready
2	Create Schemas (RAW, CLEAN, ANALYTICS)	Organize data layers, raw source data, cleaned/transformed data, and modeling layer	3 schemas created
3	Define Naming Conventions	Establish standards for database, schema, table, and column names (e.g., snake_case, upper_case)	Naming convention document
4	Document Folder Structure	Create short documentation for structure	Warehouse setup documentation

Warehouse & Database Structure

Name	Purpose	Description
BANK_WH	Act as project warehouse	Warehouse for banking churn analysis project
BANK_ANALYTICS_DB	Act as project database	Database for customer churn analysis and retention insights
BANK_ANALYTICS_DB.RAW	Act as schema for raw data	Stores raw uploaded datasets (unmodified)
BANK_ANALYTICS_DB.CLEAN	Act as schema for cleaned data	Stores cleaned and transformed datasets
BANK_ANALYTICS_DB.ANALYTICS	Act as schema for analytics purpose	Stores data models and views used by Power BI

Output & Deliverables

Deliverables	Description
Snowflake database created	BANK_ANALYTICS_DB visible in Snowflake
Schemas created	RAW, CLEAN, ANALYTICS
Warehouse setup	BANK_WH ready for compute
Documentation	steps taken documented on report

2. Load Source Data (Raw Layer)

"As a data engineer, I want to load source data into the RAW layer in Snowflake so that it's available for cleaning and analysis."

Goal: To ingest the raw customer churn dataset (bank_churn_raw.csv) into Snowflake's RAW schema, forming the foundation for data cleaning and transformation.

Tasks

#	Task	Description	Output
1	Prepare Raw Dataset File	Ensure dataset is in a clean .csv format and named consistently (e.g., bank_churn_raw.csv)	Local CSV file ready
2	Create Target Table in RAW Schema	Define a Snowflake table (RAW.BANK_CHURN_RAW) matching the dataset structure	Empty table in Snowflake
3	Stage the File	Upload the CSV file to a Snowflake *internal stage* (temporary storage area)	File staged in Snowflake
4	Load Data into Table	Use the COPY INTO command to import the staged file into the table	Data loaded successfully
5	Validate Load	Check row count, data types, and sample records	Load validated
6	Document Process	Record SQL commands, results, and screenshots	Documentation for Jira

List of files

1. Bank_churn.csv

Pre-requisites

- Snowflake database: BANK_ANALYTICS_DB
- Schemas created: RAW, CLEAN, ANALYTICS
- Warehouse: BANK_WH
- Role: DATA_ENGINEER_ROLE

Load the data (Snowsight UI)

- Go to Data > Databases > BANK_ANALYTICS_DB > RAW > Tables
- Select BANK_CHURN_RAW
- Choose Load Data → Load from file
- Upload bank_churn.csv

Actions Taken

- Created table BANK_CHURN in RAW schema
- Uploaded bank_churn.csv via Snowsight
- Verified 10,000 records loaded successfully

Validation Checks

Purpose	SQL Query
Preview First 10 records	*SELECT ** *FROM RAW.BANK_CHURN* *LIMIT 10;*
Validate Row Count	*SELECT COUNT(*) AS total_rows* *FROM RAW.BANK_CHURN; --10,000*

Results

Check	Expected	Actual	Status
Row Count	10,001	10,000 (header not included)	✅ Passed
Column Headers	14 columns	14 coluns	✅ Passed
Data Type Consistency	Matches Csv	✅ Passed	✅ Passed
Null Values	not checked		

3. Data Cleaning & Transformation

"As a data engineer, I want to extract, clean, and transform the data so that it's accurate, standardized, and ready for modelling."

Goal: Prepare the raw data for analytics — clean missing values, standardize formats, fix invalid categories, and load the cleaned dataset into the CLEAN schema.

Tasks

#	Task	Description	Output
1	Extract data from RAW	Pull data from RAW.BANK_CHURN_RAW to Python/SQL for cleaning	Dataset loaded into environment
2	Handle missing / invalid data	Fill or remove nulls, validate dates, fix mapping (e.g., GeographyID → Country)	Clean dataset
3	Standardize data formats	Convert Bank_DOJ to proper date type, ensure numeric precision	Consistent formats
4	Apply business rules	Use BRD rules (Exited=1 → churned, etc.)	Logical consistency ensured
5	Load back to Snowflake	Save cleaned data into CLEAN.BANK_CHURN_CLEANED	Clean layer table ready
6	Validate data quality	Check row count, null %, value consistency	Data quality verified

Pre-requisites

- ✅ Snowflake database: BANK_ANALYTICS_DB
- ✅ RAW.BANK_CHURN_RAW table already populated (from User Story 2)
- ✅ Access to Python / Snowflake connector
- ✅ BRD and FRD reviewed for business rules

Process Steps

a. Extract RAW data from snowflake

```
import pandas as pd
from snowflake.connector import connect
import warnings
warnings.filterwarnings('ignore')

details = {
            'user': 'NASRULKHAIR',
            'password': 'C@de0123456789',
            'account': 'SSDSCZP-MD39179',
            'warehouse': 'BANK_WH',
            'database': 'BANK_ANALYTICS_DB',
            'schema': 'RAW'
}

conn = connect(**details)

query = 'SELECT * FROM BANK_CHURN'
df  = pd.read_sql(query, conn)
df.head()
```

b. Inspect for Missing / Invalid Data

```
        df.info()
        df.isnull().sum()
        df.describe()
```

c. Clean & Transform

```
# renaming column name

df = df.rename(columns={
                'ROWNUMBER':'row_number',
                'CUSTOMERID':'customer_id',
                'CREDITSCORE':'credit_score',
                'GEOGRAPHYID': 'geography_id',
                'GENDERID': 'gender_id',
                'AGE': 'age',
                'TENURE': 'tenure',
                'BALANCE': 'balance',
                'NUMOFPRODUCTS': 'num_of_products',
                'HASCRCARD': 'has_cr_card',
                'ISACTIVEMEMBER': 'is_active_member',
                'ESTIMATEDSALARY': 'estimated_salary',
                'EXITED': 'exited',
                'BANK_DOJ': 'bank_doj'
})

df.head()
```

```
# changing bank_doj to datetime format
df['bank_doj'] = pd.to_datetime(df['bank_doj'])
print(df.dtypes)
```

```
# map geography_id to country names

df['geography_name'] = df['geography_id'].map({1: 'France', 2: 'Spain', 3:
'Germany'})
df['geography_name'].value_counts()
```

```
# map gender_id to gender

df['gender'] = df['gender_id'].map({1:'Male',2: 'Female'})
df['gender'].value_counts()
```

```
# map exited to exit and retain

df['retain_status'] = df['exited'].map({1:'Exit', 0: 'Retain'})
df['retain_status'].head()
```

```
# mapr has_cr_card to credit card holder and non credit card holder

df['credit_card_holder'] = df['has_cr_card'].map({1:'Credit Card Holder', 0: 'Non
Credit Card Holder'})
df['credit_card_holder'].value_counts()
```

```
# dropping unwanted columns
df = df.drop(columns=['row_number'])
df.head()
```

d. Load Cleaned Data to Snowflake (CLEAN Layer)

```
from snowflake.connector.pandas_tools import write_pandas

# Make sure the CLEAN schema exists
conn.cursor().execute("USE SCHEMA BANK_ANALYTICS_DB.CLEAN")

# Write dataframe to Snowflake
success, nchunks, nrows, _ = write_pandas(
        conn=conn,
        df=df,
        table_name='BANK_CHURN_CLEANED',
        schema='CLEAN',  # target schema
        auto_create_table=True,  # create if doesn't exist
        overwrite=True  # replaces existing table
)

print(f"Upload successful: {success}, rows inserted: {nrows}")
```

e. Validate Clean Load

```
SELECT *
FROM CLEAN.BANK_CHURN_CLEANED
LIMIT 5;
```

Data Quality Validation

Check	Description	Expected	Result
Row Count	Matches RAW layer	10,000	10,000
Null Values	Cleaned	0 nulls in key columns	✅
Category Mapping	Correct Geography/Gender	France/Spain/Germany; Male/Female	✅
Business Logic	"Churned" correctly derived	Consistent	✅

Output

✅ Clean table created: BANK_ANALYTICS_DB.CLEAN.BANK_CHURN_CLEANED

Completion Criteria

1. Data extracted from RAW layer
2. Cleaning rules applied per FRD
3. Data loaded into CLEAN layer successfully
4. Validation complete (row counts, data types, mappings)
5. Documented in Jira

Data Modelling & Exploration

1. Analytical & Data Model Design

Story: “As a data modeler, I want to design a star schema so that Power BI and analysts can efficiently query, aggregate, and visualize customer churn insights.”

Goal:

To create a logical and physical data model that supports churn analysis, ensuring:

- Optimized query performance for reporting tools (Power BI, SQL)
- Clear separation between fact (measures) and dimension (contextual) data
- Conformance with Snowflake dimensional modeling best practices

<u>Pre-requisites</u>

- Cleaned dataset loaded into CLEAN.BANK_CHURN_CLEANED
- Business objectives, KPIs, and churn logic defined in BRD & FRD
- Snowflake environment and schemas created (RAW, CLEAN, ANALYTICS)

<u>Data Model Design (STAR Schema)</u>

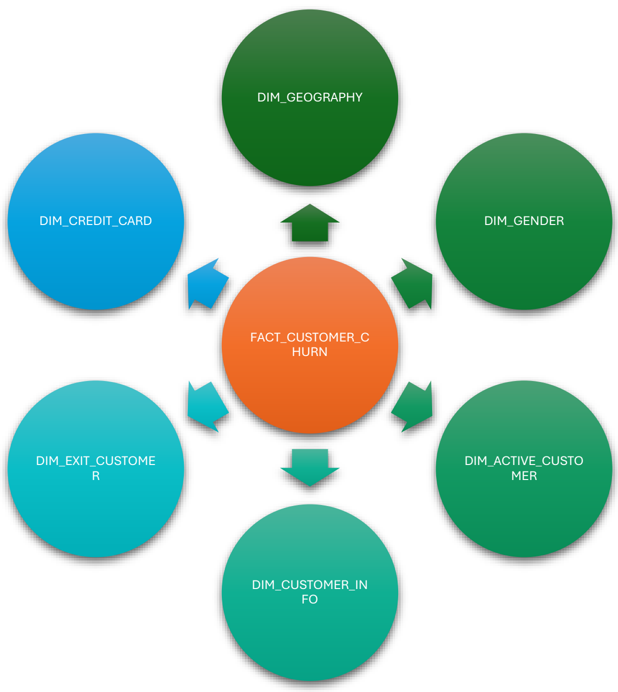


Table Definition

1. Fact_Customer_Churn_v1

Column	Data Type	Description
customer_id	INT	FK to DIM_CUSTOMER
geography_id	INT	FK to DIM_GEOGRAPHY
gender_id	INT	FK to DIM_GENDER
exit_id	INT	FK to DIM_EXIT_CUSTOMER
active_id	INT	FK to DIM_ACTIVE_CUSTOMER
cr_card_id	INT	FK to DIM_CREDIT_CARD
credit_score	INT	Customer credit score
age	INT	Customer age
tenure	INT	Years with bank
balance	FLOAT	Account balance
num_of_products	INT	Products owned
estimated_salary	FLOAT	Annual income
churned	VARCHAR	Exited, Retained
bank_doj	TIMESTAMP	Load timestamp

2. Dim_Customer

Column	Data Type	Description
customer_id	INT	PK
customer_name	VARCHAR	Name of Customer

3. Dim_Geography

Column	Data Type	Description
geography_id	INT	PK
country_name	VARCHAR	Country Name

4. Dim_Gender

Column	Data Type	Description
gender_id	INT	PK
gender	VARCHAR	Gender Label

5. Dim_Exit_Customer

Column	Data Type	Description
exit_id	INT	PK
exit_category	VARCHAR	Exit Category

6. Dim_Active_Cutomer

Column	Data Type	Description
active_id	INT	PK
active_category	VARCHAR	Active Category

7. Dim_Credit_Card

Column	Data Type	Description
credit_id	INT	PK
Category	VARCHAR	Credit Crad label

SQL Implementation

1. Tables Modelling

```
-- Dimension Tables

-- 1. DIM_GEOGRAPHY

CREATE OR REPLACE TABLE DIM_GEOGRAPHY (
geography_id INT PRIMARY KEY,
country_name VARCHAR
);

-- 2. DIM_CUSTOMER

CREATE OR REPLACE TABLE DIM_CUSTOMER (
customer_id INT PRIMARY KEY,
customer_name VARCHAR
);

-- 3. DIM_GENDER

CREATE OR REPLACE TABLE DIM_GENDER (
gender_id INT PRIMARY KEY,
gender VARCHAR
);

-- 4. DIM_EXIT_CUSTOMER

CREATE OR REPLACE TABLE DIM_EXIT_CUSTOMER (
exit_id INT PRIMARY KEY,
exit_category VARCHAR
);

-- 5. DIM_ACTIVE_CUSTOMER

CREATE OR REPLACE TABLE DIM_ACTIVE_CUSTOMER (
active_id INT PRIMARY KEY,
active_category VARCHAR
);

-- 6. DIM_CREDIT_CARD

CREATE OR REPLACE TABLE DIM_CREDIT_CARD (
credit_id INT PRIMARY KEY,
category VARCHAR
);

-- FACT_TABLE

CREATE OR REPLACE TABLE FACT_CUSTOMER_CHURN_v1 AS
SELECT *
FROM CLEAN.BANK_CHURN_CLEANED;
```

2. Adding FK Constraints

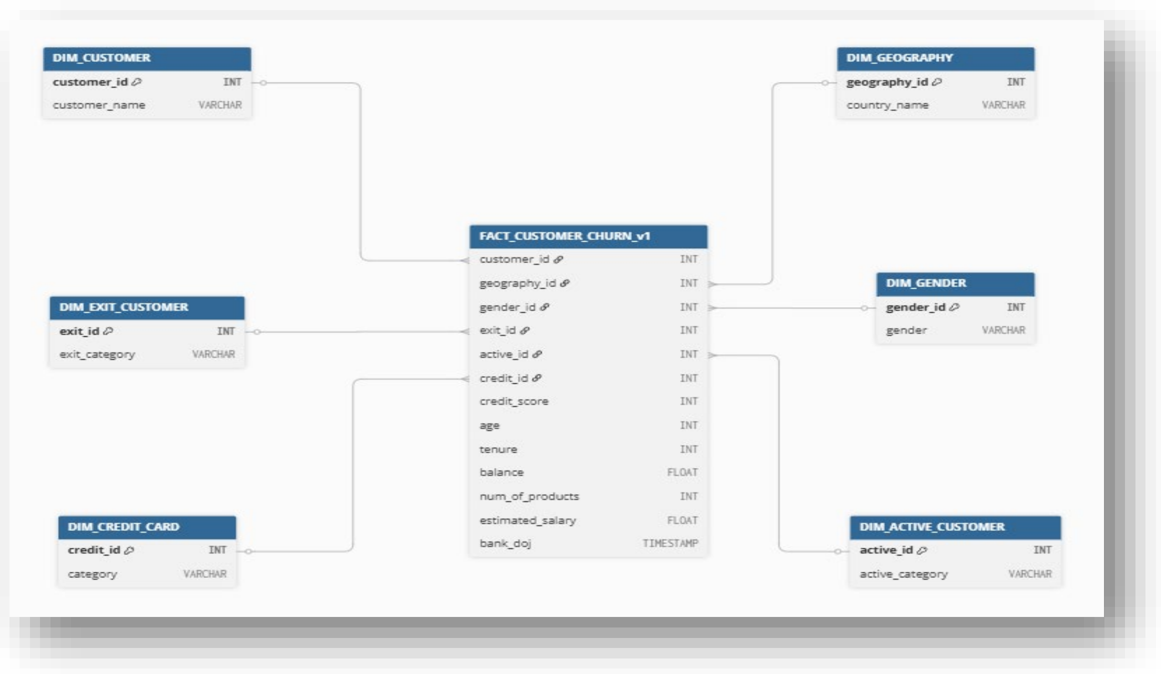
```
-- UPDATING FK to PK each tables

ALTER TABLE FACT_CUSTOMER_CHURN_V1
ADD CONSTRAINT fk_customer
FOREIGN KEY ("customer_id") REFERENCES ANALYTICS.DIM_CUSTOMER(customer_id);

ALTER TABLE FACT_CUSTOMER_CHURN_V1
ADD CONSTRAINT fk_geography
FOREIGN KEY ("geography_id") REFERENCES ANALYTICS.DIM_GEOGRAPHY(geography_id);


ALTER TABLE FACT_CUSTOMER_CHURN_V1
ADD CONSTRAINT fk_gender
FOREIGN KEY ("gender_id") REFERENCES ANALYTICS.DIM_GENDER(gender_id);


ALTER TABLE FACT_CUSTOMER_CHURN_V1
ADD CONSTRAINT fk_exit_customer
FOREIGN KEY ("exit_id") REFERENCES ANALYTICS.DIM_EXIT_CUSTOMER(exit_id);

ALTER TABLE FACT_CUSTOMER_CHURN_V1
ADD CONSTRAINT fk_active_customer
FOREIGN KEY ("active_id") REFERENCES ANALYTICS.DIM_ACTIVE_CUSTOMER(active_id);

ALTER TABLE FACT_CUSTOMER_CHURN_V1
ADD CONSTRAINT fk_credit_card
FOREIGN KEY ("cr_card_id") REFERENCES ANALYTICS.DIM_CREDIT_CARD(credit_id);
```

3. Update Records into Tables

```
-- UPDATE DATA INTO Dimension Tables

--1 . DIM_CUSTOMER

INSERT INTO DIM_CUSTOMER (customer_id, customer_name)
SELECT DISTINCT CUSTOMERID, SURNAME
FROM CLEAN.CUSTOMERINFO;

-- Validate the records

SELECT COUNT(*)
FROM DIM_CUSTOMER


-- 2. DIM_GEOGRAPHY

INSERT INTO DIM_GEOGRAPHY(geography_id, country_name)
SELECT DISTINCT "geography_id", "country_name"
FROM FACT_CUSTOMER_CHURN_V1;

-- Validate the records
SELECT DISTINCT *
FROM dim_geography;


-- 3. DIM_GENDER
INSERT INTO DIM_GENDER(gender_id, gender)
SELECT DISTINCT "gender_id", "gender"
FROM fact_customer_churn_v1;

-- Validate the records
SELECT *
FROM DIM_GENDER;

-- 4. DIM_EXIT_CUSTOMER
INSERT INTO DIM_EXIT_CUSTOMER(exit_id, exit_category)
SELECT DISTINCT "exit_id", "churned"
FROM fact_customer_churn_v1;
```

```
-- Validate the records
SELECT *
FROM DIM_EXIT_CUSTOMER;

-- 5. DIM_ACTIVE_CUSTOMER
INSERT INTO DIM_ACTIVE_CUSTOMER (active_id, active_category)
SELECT DISTINCT "active_id", "active_member"
FROM fact_customer_churn_v1;

SELECT *
FROM DIM_ACTIVE_CUSTOMER;


-- 6. DIM_CREDIT_CARD
INSERT INTO DIM_CREDIT_CARD(credit_id, category)
SELECT DISTINCT "cr_card_id", "credit_card_holder"
FROM fact_customer_churn_v1;

-- Validate the records
SELECT *
FROM DIM_ACTIVE_CUSTOMER;
```

ER DIAGRAM



Acceptance Criteria

- Star schema created successfully in ANALYTICS schema
- Foreign keys correctly reference dimension tables
- Derived columns (age group, tenure band) verified
- Row counts validated with source (CLEAN layer)
- ER diagram and documentation uploaded

2. Exploratory Data Analysis (EDA)

Story: “As a data analyst, I want to perform exploratory analysis on the cleaned data so that I can understand patterns, relationships, and factors contributing to customer churn.”

Goal:

To analyze the CLEAN.BANK_CHURN_CLEANED dataset to:

- Understand the distribution of key variables
- Explore correlations and patterns that drive churn
- Generate insights that validate data model integrity and guide dashboard design

Pre-requisites

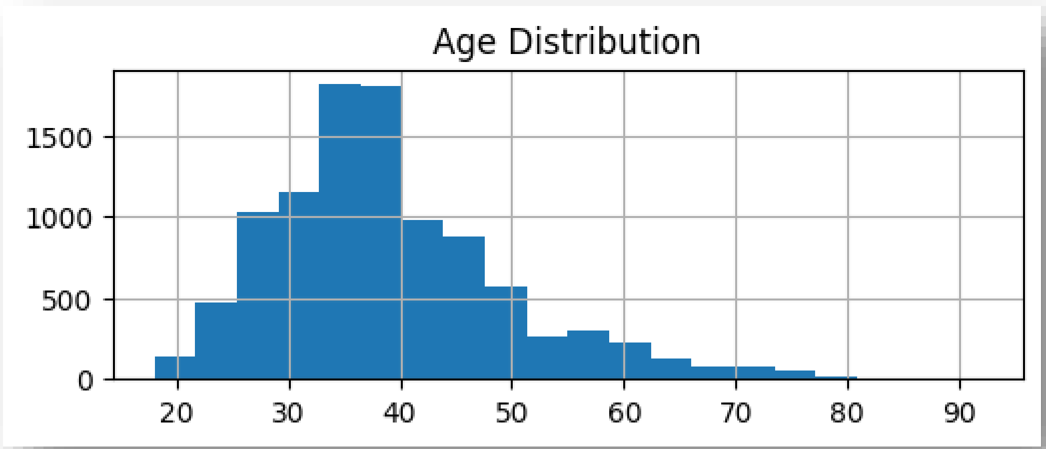
- Data model structure defined (Epic 3, Story 1)
- Cleaned data available in Snowflake (CLEAN.BANK_CHURN_CLEANED)
- Python or SQL environment ready for analysis (e.g., Jupyter, VSCode)
- Business KPIs and churn logic defined in BRD/FRD

Objectives

- Understand data distributions and detect anomalies.
- Identify patterns and relationships influencing customer churn.
- Validate data integrity post-cleaning.
- Derive actionable insights for Power BI dashboard design.

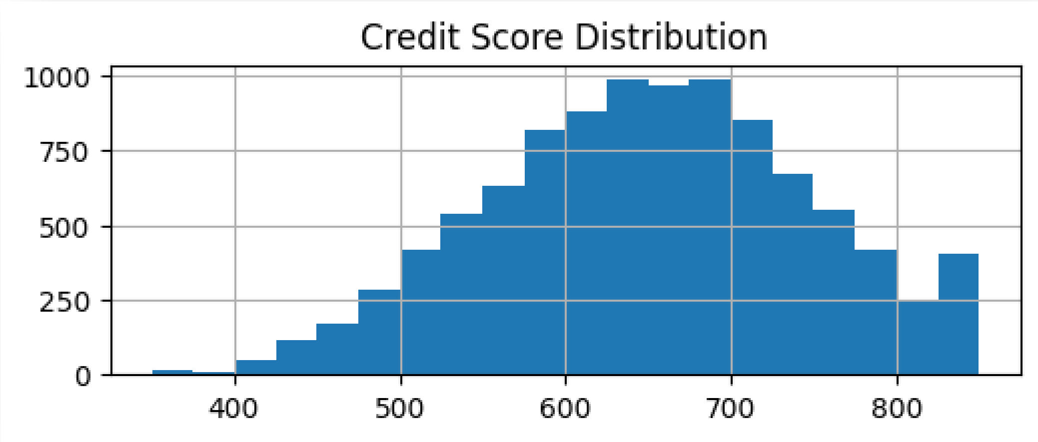
Key Analyses and Findings

1. Univariate Analysis
   a) a. Age Distribution



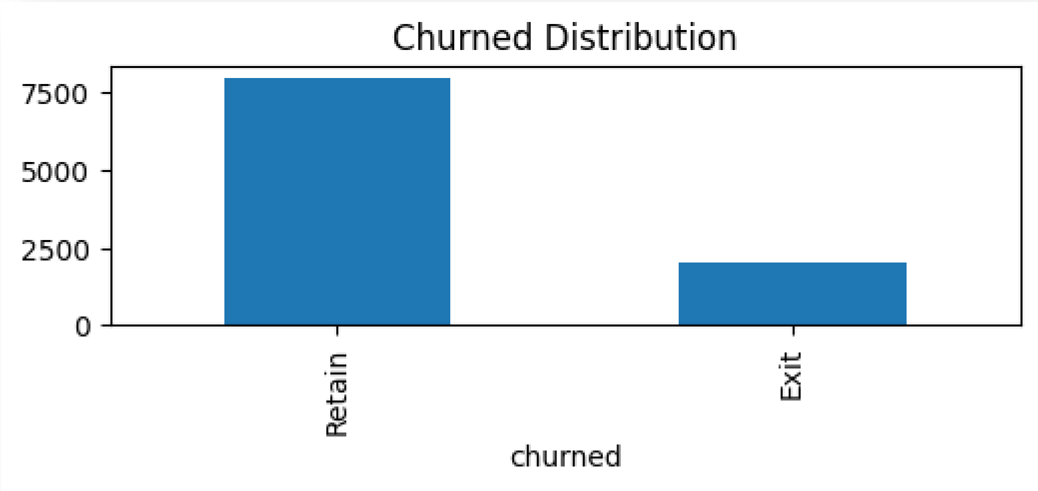
* Majority of customers are aged 35-40 years old

b) Credit Score Distribution



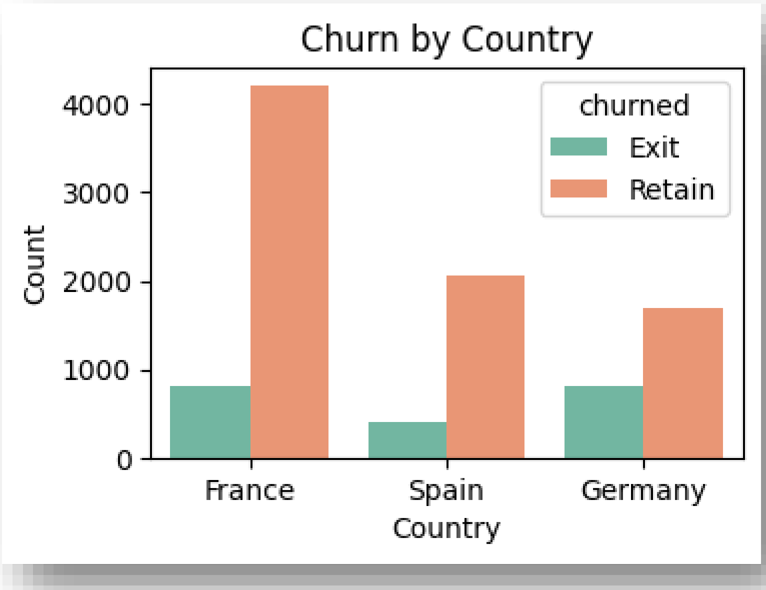
* Credit Scores cluster around 600-700 range

c) Churned Distribution

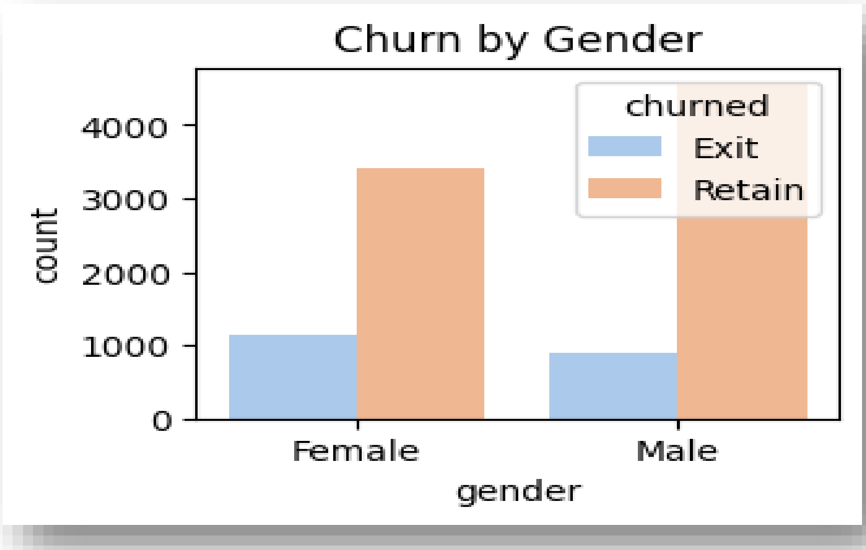


* 2,000 out of 9,500 customers exited, your churn rate is around 21%.

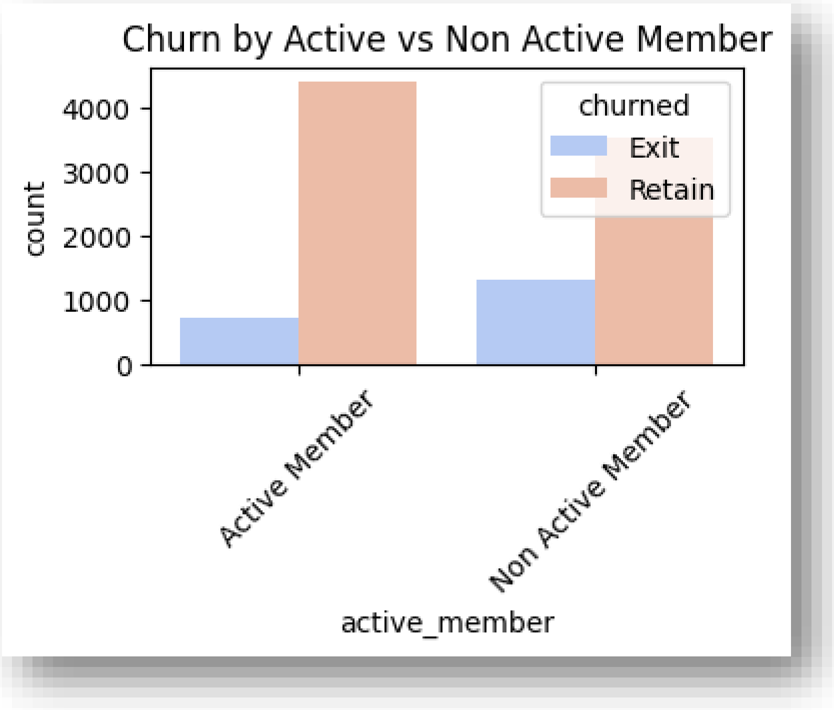
2. Bivariate Analysis
    1. Churn By Geography



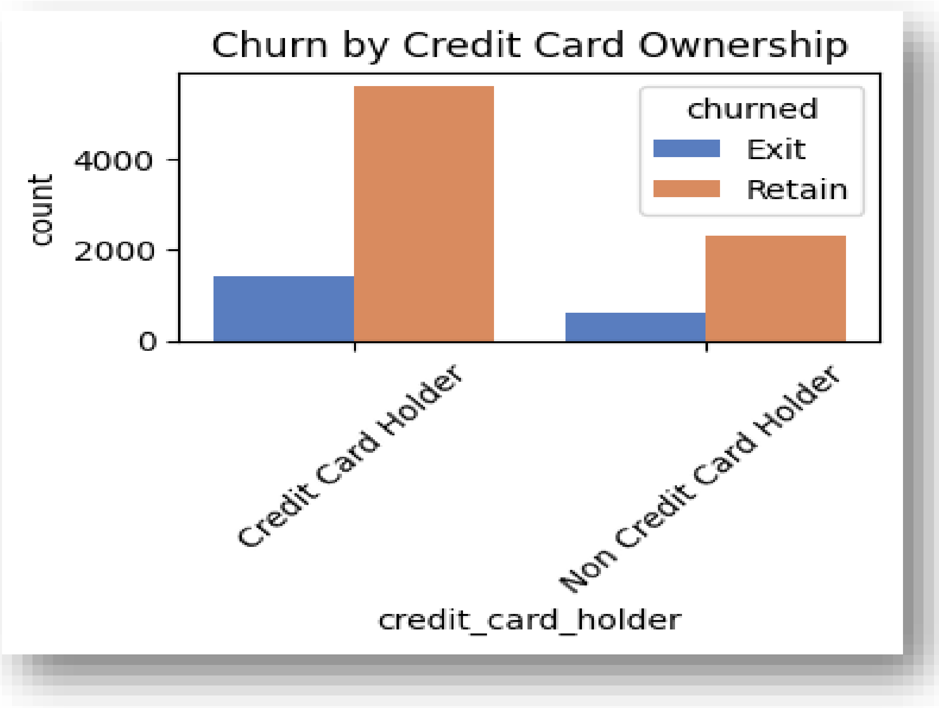
    2. Churn By Gender



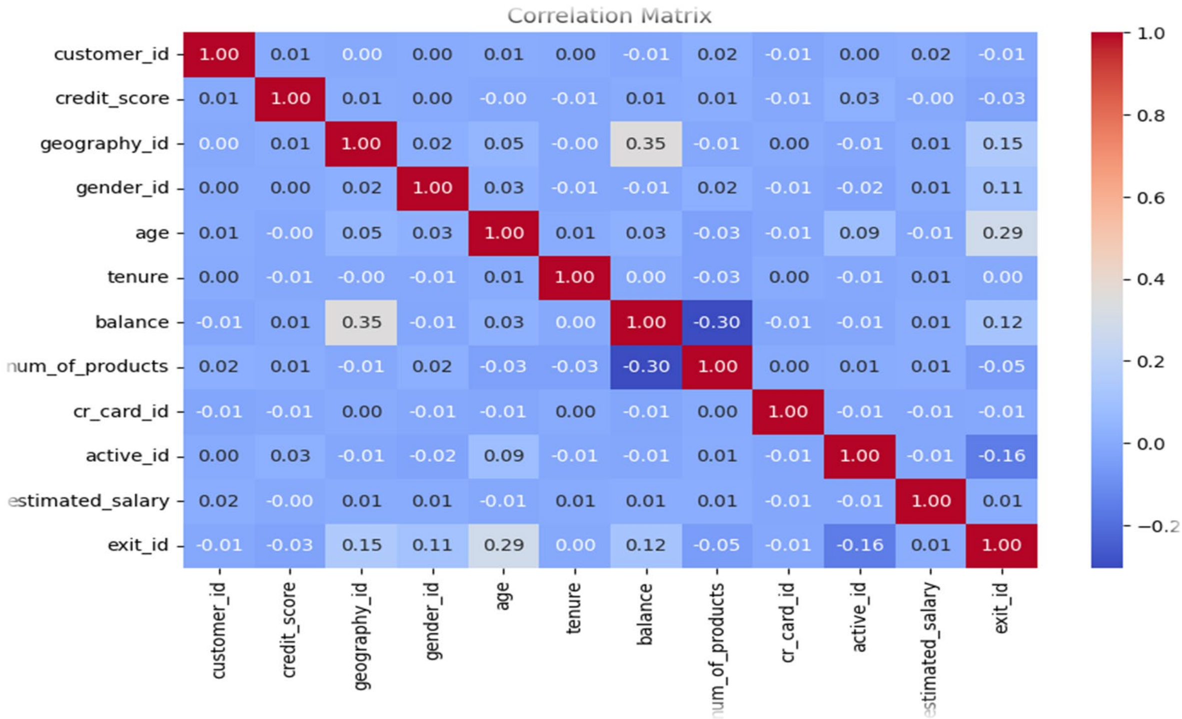
3. Churn by Active Member Status



4. Churn by Credit Card Holder Status



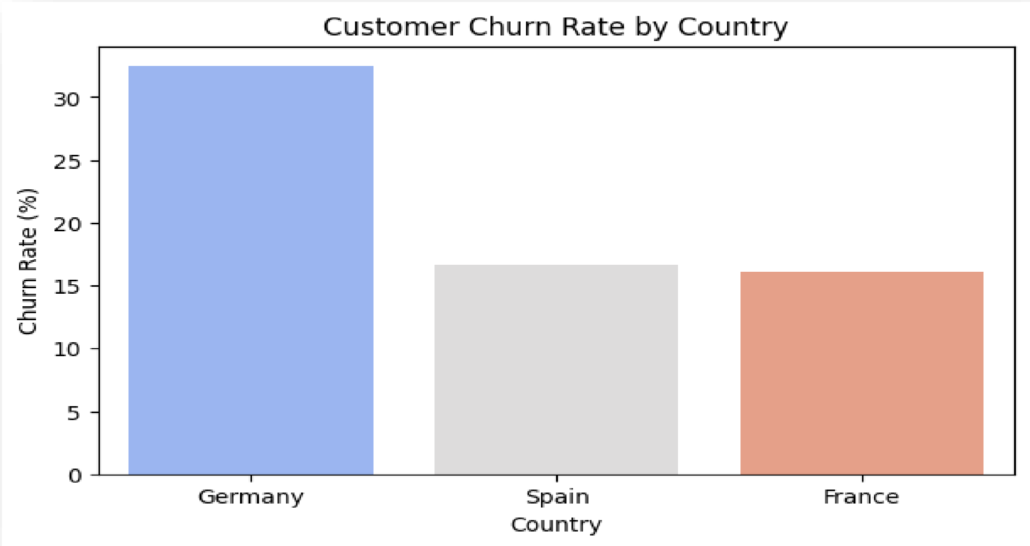
3. Correlation Matrix



Metric	Score	Correlation
Exit ID	1.000000	Irrelevant identifier
Age	0.285323	Older customers are slightly more likely to churn. Possibly due to reduced product engagement or preference for competitor services.
Geography	0.153771	Indicates churn behavior differs by country — certain regions (e.g., Spain vs. France) may have higher exit rates. Worth segmenting further.
Account Balance	0.118533	Customers with higher account balances show a slightly higher churn tendency — perhaps they're switching to competitors for better interest rates or benefits.
Gender	0.106512	Small but notable — female customers may churn slightly more. Further segmentation (age × gender) could confirm this trend.
Estimated Salary	0.012097	Salary has almost no effect on churn — income alone doesn't determine exit likelihood.
Tenure	0.000699	Loyalty duration has no clear correlation, suggesting that long-time customers are just as likely to churn as newer ones.
Customer ID	-0.006248	Irrelevant identifier
Credit Card Ownership	-0.007138	Very weak, but credit card holders slightly less likely to churn, possibly due to product stickiness.
credit_score	-0.027094	Weak — higher credit scores marginally reduce churn, suggesting financially responsible customers stay longer.
Number Of Products	-0.047820	Weak negative — customers with multiple products (e.g., savings + credit card) are slightly more loyal.
Active Member	-0.156128	Moderate negative — active customers are significantly less likely to churn, showing engagement is key for retention.

4. Segmentation Analysis

country_name	Total_Customers	Churned_Customers	Churn_Rate_%
Germany	2509	814	32.44
Spain	2477	413	16.67
France	5014	810	16.15



* Insights

- Germany shows the highest churn rate (32.4%), almost double that of France and Spain.
- France has the largest customer base (5,014 customers) but maintains the lowest churn rate (16.15%)
- Spain sits in between — moderate churn (16.7%) with a slightly smaller customer base than France.

3. Core Business Questions

Story: As a data analyst, I want to answer core business questions using SQL so that I can validate business KPIs using the cleaned data.

Goal: Design the analytical model for business analysis.

Objective

To write SQL queries in Snowflake (or locally in your environment) that answer the key business questions defined earlier in your BRD/FRD — such as:

- What is the churn rate overall and by customer segments?

- Which factors most influence churn?
- How are demographics (age, gender, geography) related to churn?
- What is the average customer profile for churned vs retained customers?

Task Breakdown

Task	Description	Deliverable
Write SQL for key business metrics	Use CLEAN.BANK_CHURN_CLEANED dataset	Reusable SQL scripts
Validate KPI logic	Check query results for accuracy	Summary KPI report
Store SQL scripts in Git	Version control for analytics queries	Organized repo folder `/sql/business_metrics`

Core Business Metrics Definition

KPI Name	Formula	Business Meaning
Churn Rate	SUM(Exited)* 100/COUNT(*)	% of customers who left
Average Credit Score (Churned vs Retained)	AVG(CreditScore)	Credit profile by churn group
Average Tenure by Churn Group	AVG(Tenure)	How long churned customers stayed
Churn Rate by Country	Grouped churn %	Identify high-risk regions
Churn Rate by Product Holding	Grouped churn % by NumOfProducts	Cross-sell opportunity

Business Questions (Part 1)

1. Overall Churn Rate

Sql

```
SELECT
  ROUND(SUM("exit_id") * 100 / COUNT(*),2) AS churn_rate_percentage
FROM BANK_CHURN_CLEANED;
```

Results

CHURN_RATE_PERCENTAGE
20.37

2. Churn Rate by Country

Sql

```
SELECT
  "country_name",
  ROUND(SUM("exit_id") * 100 / COUNT(*),2) AS churn_rate_percentage
FROM BANK_CHURN_CLEANED
GROUP BY ALL;
```

Results

country_name	CHURN_RATE_PERCENTAGE
France	16.15
Spain	16.67
Germany	32.44

3. Churn Rate by Gender

Sql	SELECT "gender", ROUND(SUM("exit_id") *100 / COUNT(*), 2) AS churn_rate_percentage FROM bank_churn_cleaned GROUP BY ALL;
Results	

gender	CHURN_RATE_PERCENTAGE
Female	25.07
Male	16.46

4. Churn Rate by Age Group

Sql

```
SELECT
  CASE
    WHEN "age" < 30 THEN 'Under 30'
    WHEN "age" BETWEEN 30 AND 39 THEN '30s'
    WHEN "age" BETWEEN 40 AND 49 THEN '40s'
    WHEN "age" BETWEEN 50 AND 59 THEN '50s'
    ELSE '60s +'
  END AS age_group,
  ROUND(SUM("exit_id") * 100 / COUNT(*), 2) AS churn_rate_percentage
FROM BANK_CHURN_CLEANED
GROUP BY ALL
ORDER BY churn_rate_percentage DESC;
```

Results

AGE_GROUP	CHURN_RATE_PERCENTAGE
50s	56.04
40s	30.79
60s +	27.95
30s	10.88
Under 30	7.56

5. Churn Rate by Product Holding

Sql

```
SELECT
  "num_of_products" AS product_holding,
  ROUND(SUM("exit_id") * 100 / COUNT(*), 2) churn_rate_percentage
FROM BANK_CHURN_CLEANED
GROUP BY ALL
ORDER BY product_holding ASC;
```

Results

PRODUCT_HOLDING	CHURN_RATE_PERCENTAGE
1	27.71
2	7.58
3	82.71
4	100.00

6. Average Credit Score and Balance (Churned vs Retained)

Sql	SELECT "churned" AS churn_status, ROUND(AVG("credit_score"), 2) AS avg_credit_score, ROUND(AVG("balance"), 2) AS avg_account_balance, FROM BANK_CHURN_CLEANED GROUP BY ALL;
Results	

CHURN_STATUS	AVG_CREDIT_SCORE	AVG_ACCOUNT_BALANCE
Exit	645.35	91108.54
Retain	651.85	72745.3

7. Average Tenure and Age (Churned vs Retained)

Sql	SELECT "churned" AS churn_status, ROUND(AVG("tenure"), 2) AS avg_tenure, ROUND(AVG("age"), 0) AS avg_age FROM CLEAN.BANK_CHURN_CLEANED GROUP BY ALL;
Results	

CHURN_STATUS	AVG_TENURE	AVG_AGE
Retain	4.86	37
Exit	4.87	45

Summarize Findings

KPI	Insights
Churn Rate	~20% of customers churned
Country	Germany has the highest churn (≈32%)
Gender	Slightly higher churn among females
Age	Older customers (40+) show more attrition
Activity	Inactive customers churn 3× more
Product Holding	The highest number of products, the higher churn rate

Business Question (Part 2)

1. Top 10 Customers with Highest Churn Risk

Sql

```
/*
credit_score: weight= 0.4 (1st)
balance: weight= 0.3 (2nd)
tenure: weight= 0.2 (3rd)
activer_member: weight = 0.1 (4th)
*/


WITH churn_score AS (
  SELECT
    bc."customer_id",
    ci."SURNAME",
    bc."country_name",
```

	`` bc."gender", bc."age", bc."credit_score", bc."balance", bc."tenure", bc."active_member", ROUND((0.4 * (1 - bc."credit_score" / 1000.0)) + (0.3 * (bc."balance" / (SELECT MAX("balance") FROM bank_churn_cleaned))) + (0.2 * (bc."tenure" / (SELECT MAX("tenure") FROM bank_churn_cleaned))) + (0.1 * (CASE WHEN bc."active_member" = 'Active Member' THEN 0 ELSE 1 END)) , 2) AS churn_risk_score FROM BANK_CHURN_CLEANED bc JOIN CUSTOMERINFO ci ON bc."customer_id" = ci."CUSTOMERID" WHERE bc."churned" = 'Retain'), ranked AS (SELECT *, DENSE_RANK() OVER (ORDER BY churn_risk_score DESC) AS customer_risk_rank FROM churn_score) SELECT * FROM ranked WHERE customer_risk_rank <= 10 ORDER BY customer_risk_rank ASC; ```
Results	**refer csv table (Top 10 High Risk) stored in Github*

2. Churn Trend By Year

Sql

```
SELECT
  EXTRACT(YEAR FROM "BANK_DOJ") AS year,
  COUNT(*) AS total_customers,
  SUM("exit_id") AS total_customers_churned,
  ROUND(SUM("exit_id") * 100 / COUNT(*), 2) AS churn_rate_percentage
FROM bank_churn_cleaned
GROUP BY ALL
ORDER BY year DESC;
```

Results

YEAR	TOTAL_CUSTOMERS	TOTAL_CUSTOMERS_CHURNED	CHURN_RATE_PERCENTAGE
2019	3302	655	19.84
2018	2591	523	20.19
2017	2137	478	22.37
2016	1965	381	19.39
2015	5	0	0.00

3. Rank Countries by Churn Rate and Average Credit_score

Sql

```
WITH churn_rate AS(
  SELECT
    "country_name",
    ROUND(AVG("credit_score"), 2) AS avg_credit_score,
    ROUND(SUM("exit_id") *100 / COUNT(*),2) AS churn_rate_percentage
  FROM bank_churn_cleaned
  GROUP BY ALL
)
```

	SELECT *, RANK() OVER(ORDER BY churn_rate_percentage DESC) churn_rank FROM churn_rate;
Results	

country_name	AVG_CREDIT_SCORE	CHURN_RATE_PERCENTAGE	CHURN_RANK
Germany	651.45	32.44	1
Spain	651.33	16.67	2
France	649.67	16.15	3

4. Gender-Based Churn Analysis per Country

Sql

```
SELECT
  "country_name",
  "gender",
  SUM(CASE WHEN "churned" = 'Exit' THEN 1 ELSE 0 END) AS total_churn,
  ROUND(SUM(CASE WHEN "churned" = 'Exit' THEN 1 ELSE 0 END) * 100 / COUNT(*), 2)
churn_rate_percentage
FROM bank_churn_cleaned
GROUP BY ALL;
```

Results

country_name	gender	TOTAL_CHURN	CHURN_RATE_PERCENTAGE
France	Male	350	12.71
France	Female	460	20.34
Spain	Female	231	21.21
Germany	Female	448	37.55
Germany	Male	366	27.81
Spain	Male	182	13.11

5. Loyal Customers (Active, Long Tenure (>5 years), No Churn)

Sql

```
SELECT
  ci."SURNAME",
  bc."country_name",
  bc."age",
  bc."balance",
  bc."estimated_salary"
FROM BANK_CHURN_CLEANED bc
JOIN CUSTOMERINFO ci
ON bc."customer_id" = ci.customerid
WHERE bc."tenure" >= 5
  AND bc."active_member" = 'Active Member'
  AND bc."churned" = 'Retain'
ORDER BY bc."country_name", bc."estimated_salary" DESC;
```

Results **refer csv table (Top 10 High Risk) stored in Github*

6. Correlation Between Age & Balance

Sql

```
SELECT
 ROUND(
  (COUNT(*) * SUM("age" * "balance") - SUM("age") * SUM("balance")) /
  SQRT(
   (COUNT(*) * SUM(POWER("age", 2)) - POWER(SUM("age"), 2)) *
```

	(COUNT(*) * SUM(POWER("balance", 2)) - POWER(SUM("balance"), 2))), 3) AS correlation_age_balance FROM BANK_CHURN_CLEANED WHERE "churned" = 'Exit';
Results	CORRELATION_AGE_BALANCE -0.023

7. Average Salary & Balance by Product Count and Churn Status

Sql	SELECT "num_of_products", "churned", ROUND(AVG("estimated_salary"), 2) AS avg_salary, ROUND(AVG("balance"), 2) AS avg_balance, COUNT(*) AS total_customers FROM BANK_CHURN_CLEANED GROUP BY ALL ORDER BY "num_of_products", "churned";
Results	(see table below)

num_of_products	churned	AVG_SALARY	AVG_BALANCE	TOTAL_CUSTOMERS
1	Exit	100639.83	92028.82	1409
1	Retain	99045.36	101052.82	3675
2	Exit	100883.31	90252.36	348
2	Retain	100416.63	48731.13	4242
3	Exit	106776.61	85853.09	220
3	Retain	92560.21	25744.26	46
4	Exit	104763.72	93733.14	60

Summarize Findings

KPI	Insights
Top 10 High-Risk Customers	Non-active customers with low credit scores (below 560) and high balances (≈180K+) show the highest churn risk, especially in Germany and France.
Churn Trend by Year	Churn rate peaked in 2017 (22.4%) and gradually declined to around 19–20% in recent years, suggesting improved customer retention since 2018.
Country Churn Ranking	**Germany** records the **highest churn rate (32.4%)** despite having the **highest average credit score**, indicating that churn there is likely driven by **non-financial factors** such as customer experience or market competition.
Gender-Based Churn by Country	Female customers show consistently higher churn rates across all countries, especially in Germany (37.6%), suggesting a potential gap in customer satisfaction or engagement among female clients
Loyal Customers (Active & Long Tenure)	Out of 2,479 loyal customers, the majority are active members from France with tenure over 5 years and high estimated salaries (≈200K), reflecting strong retention among financially stable, long-term clients.

Correlation Between Age & Balance (Churned Customers)	The correlation coefficient of -0.023 indicates no significant relationship between age and account balance among churned customers — churn behavior appears independent of age-related financial balance.
Average Salary & Balance by Product Count	Customers holding 3–4 products show the highest churn rates with higher balances (≈85K–94K), indicating that overextension or product complexity may contribute to churn despite higher income levels.

Insights & Recommendations

Insight	Recommendation
Germany shows highest churn	Investigate customer satisfaction or competitive offers in Germany
Older customers (40–50) leaving more	Offer long-term loyalty or tenure-based benefits
Inactive members more likely to churn	Launch re-engagement campaigns for inactive users
1-product customers have high churn	Encourage cross-selling (credit card, savings)
Credit score and balance moderately related to churn	Introduce financial wellness programs or rewards
Non-active customers with low credit scores (below 560) and high balances (≈180K+) show the highest churn risk, especially in Germany and France.	Implement targeted retention campaigns for high-value inactive customers, offering personalized financial advice or exclusive account perks to re-engage them.
Churn rate peaked in 2017 (22.4%) and improved to ~19–20% after 2018, showing better customer retention over time.	Continue current retention strategies and analyze what post-2018 improvements (e.g., policy, product, or service changes) helped reduce churn to reinforce them further.
Germany records the highest churn (32.4%) despite high credit scores, indicating non-financial reasons for attrition.	Conduct customer satisfaction surveys and competitor benchmarking in Germany to uncover service or experience gaps driving attrition.
Female customers show higher churn rates across all countries, especially Germany (37.6%).	Introduce female-focused engagement initiatives, such as personalized offers, community programs, or financial education content.
Out of 2,479 loyal customers, most are active members from France with long tenure and high salaries (~200K).	Strengthen loyalty programs by offering premium-tier rewards and referral benefits for long-term active customers to maintain satisfaction and advocacy.
Weak correlation (-0.023) between age and balance among churned customers indicates churn is not driven by age or wealth.	Focus churn prevention on behavioral and engagement factors (e.g., inactivity, low product usage) rather than demographics.
Customers holding 3–4 products have the highest churn despite higher balances.	Simplify multi-product management and offer bundled product discounts or financial guidance to reduce confusion and product fatigue.

Outcome

- SQL queries provide complete visibility into customer churn behavior
- Core business KPIs validated and ready for visualization
- Analytical results will now feed into **Power BI data model (Epic 4)**

Visualization & Power BI Development

1. Snowflake – Power BI Connection

Story: “As a report developer, I want to connect Power BI to Snowflake so that I can visualize live, cleaned data for analysis and reporting.”

Goal: Build powerful, interactive Power BI reports and dashboards that bring the customer churn analysis to life.

Objective:

To establish a secure and reliable connection between Power BI Desktop and Snowflake, allowing data analysts to build and publish reports based on the cleaned data (CLEAN schema) within Snowflake.

Prerequisites

Requirements	Details
Power BI Desktop	Installed (latest version recommended)
Snowflake Account	Active with necessary privileges
Snowflake ODBC Driver	Installed and configured
Access Role	Read access to Analytics schema
Workspace	Power BI workspace created

Data Source Setup

Parameter	Value
Warehouse	BANK_ANALYTICS_WH
Database	BANK_ANALYTICS_DB
Schema	ANALYTICS
Tables(s)	• DIM_ACTIVE_CUSTOMER *(Import)* • DIM_CREDIT_CARD *(Import)* • DIM_CUSTOMER *(Import)* • DIM_EXIT_CUSTOMER *(Import)* • DIM_GENDER *(Import)* • DIM_GEOGRAPHY *(Import)* • FACT_CUTOMER_CHURN_V1 *(Direct Query)*
Connection Type	Import (for stable refresh) or DirectQuery (for live updates)
Authentication	Username & Password or SSO

Steps to Connect Power BI to Snowflake

1. Launch Power BI Desktop
2. Snowflake Connection Details
3. Authentication
4. Navigator Window
5. Data Preview & Validation
6. Save Power BI File

Direct Query Fact Table (M Language)

Table	M Language
FACT_CUSTOMER_CHURN_V1	*= Sql.Database("NASRULKHAIR\SQLEXPRESS", "Data Warehouse", [Query = " SELECT * FROM dbo.Fact_Bank_Churn "])*

Data Validation Checklist

Validation	Expected	Status
Row Count	10,000	✅
Column Count	13	✅
Data Type Consistency	Numeric / Date / Binary / Varchar verified	✅
Null / Blank	No Tolerance	✅
Sample Records	Match between Snowflake and Power BI preview	✅

Deliverables

Deliverable	Description
Bank_Churn_Analysis.pbix	Power BI report file with Snowflake connection
Data Connection Screenshot	Visual proof of successful setup
Validation Log	Record of row/column verification
Connection Doc	This documentation

Outcome

- Power BI successfully connected to Snowflake CLEAN schema
- Dataset validated for accuracy and consistency
- Ready for report development and DAX measures (next story)

2. Data Enrichment

Story: As a data analyst, I want to create derived columns (like age group, tenure level, and balance segment) so I can analyze customer behavior in meaningful categories.

Goal: To enhance the existing Power BI data model by creating new derived fields, lookup tables, and behavioral metrics that allow advanced segmentation, trend analysis, and churn insights, without altering the original database structure.

Objective

- Establish an enriched data model in Power BI using Power Query and DAX.
- The goal is to add calculated dimensions (e.g., Age Group, Tenure Level, Credit Score Band), integrate new lookup tables, and develop customer value metrics for more impactful reporting.

Prerequisites

Requirements	Details
Power Bi Desktop	Installed (latest version recommended)
Existing Model	Base model connected to Snowflake (FACT_CUSTOMER_CHURN_V1 + DIM tables)
Access Role	Analyst-level permission to modify Power BI data model
Documentation	Latest ERD and data dictionary from previous epic
Workspace	Power BI workspace created for enrichment update

Data Source Setup

Table	Type	Connection Mode	Purpose
FACT_CUSTOMER_CHURN_V1	Fact	Direct Quey / Import	Base table for churn metrics
DIM_ACTIVE_CUSTOMER	Dimension	Import	Lookup for activity category
DIM_CREDIT_CARD	Dimension	Import	Lookup for credit card ownership
DIM_CUSTOMER	Dimension	Import	Lookup for customer names
DIM_EXIT_CUSTOMER	Dimension	Import	Lookup for churn status
DIM_GENDER	Dimension	Import	Gender reference
DIM_GENDER	Dimension	Import	Country reference
New: DIM_REASON	Dimension	Derived	Churn reason reference
New: DIM_TITLE	Dimension	Derived	Reporting Visual reference
New: DIM_CREDIT_SCORE_BAND	Dimension	Derived	Credit Score reference
New: DIM_DATE			Time Intelligence reference

Enrichment Components

Feature	Type	Logic/Formula	Purpose
age_group	calculated column	IF Age < 30 → "Young", 30–40 → "Mid-Age", 41–50 → "Mature", >50 → "Senior"	
tenure_group	calculated column	IF Tenure <= 3 → "New", 4–6 → "Mid Term", >6 → "Long Term"	
balance_flag	calculated column	BINS by percentile ("Has Balance" / "No Balance")	
score_band	dimension table	CASE: <600 = Poor, 600–700 = Fair, 701–800 = Good, >800 = Excellent	
churn_reason	dimension table		
balance_flag	calculated column		

3. Analytical Report

Story: "As a report developer, I want to create analytical Power BI reports and DAX-based KPIs so that business users can easily understand customer churn patterns and performance metrics."

Goal: To design and develop a visually compelling Power BI dashboard

Objective

To design and develop a visually compelling Power BI dashboard that summarizes customer churn insights. This includes:

- Defining DAX measures for KPIs
- Creating visualizations for churn trends, demographics, and customer behaviour
- Organizing data into meaningful report pages

Data Model Overview

Fact Table:

- FACT_CUSTOMER_CHURN_V1: Central transaction and behavioural dataset

Dimension Table:

- DIM_CUSTOMER: Customer identity (fk - customer_id)

- DIM_GEOGRAPHY: Location-based grouping (France, Spain, Germany)
- DIM_GENDER: Gender segmentation
- DIM_REASON: Exit or Churn reason categories
- DIM_CREDIT_CARD: Credit card holder classification
- DIM_ACTIVE_CUSTOMER: Active status
- DIM_EXIT_CUSTOMER: Retain or Exit flag
- DIM_DATE: For temporal analysis

Key Metrics & KPISs

KPI	Purpose
Total Customers	Shows the total active customers
Total Churned Customers	Total number of churned customers
Churn Rate (%)	Percentage of customers lost
Avg. Tenure	Average duration before churn
Retention Rate (%)	Shows loyalty trend
Avg. Credit Score (Churned vs Retained)	Credit quality of churned customers
Avg. Credit Score (Churned vs Retained)	Credit quality of churned customers
Average Tenure (Years)	Customer lifetime duration
Active Customers	Currently engaged customers
Credit Card Holders (%)	Product penetration
Avg Estimated Salary	Demographic financial capability
Churn by Credit Card	Impact of credit card ownership
Churn by Geography	Regional churn insight
Churn Reason Count	Categorize churn motives
Churn Impact Index (Advanced)	Financial churn impact estimate
Customer Loyalty Score (Advanced)	Composite loyalty strength
Revenue Potential (Simulated)	Simulated business revenue potential

Advanced Visuals & Analytical Techniques

Technique	Description
Decomposition Tree	Drill into churn reasons dynamically (Reason → Gender → Geography → Credit Card)
Key Influencers	Automatically detect which features contribute most to churn (e.g., Age, Credit Score, Geography)
Smart Narrative	Summarize story textually (“Churn rate in Germany is 10% higher than average…”)
What-If Parameter	Add slider to simulate churn reduction and revenue impact
Dynamic Segmentation	Group customers dynamically: *High Risk, Medium Risk, Low Risk* based on DAX
Bookmark Navigation	Interactive storytelling through toggled views (Overview ↔ Reason Explorer)

Dashboard Design & Layout

1. Executive Overview

Objective

Provide a high-level summary of customer churn performance and key retention indicators. This page acts as the management landing page, giving executives a snapshot of customer count, churn trend, retention rate, and regional distribution with drill-through options to deeper insights.

Overview

The Executive Overview provides a summary of customer churn performance between 2016 and 2019, highlighting total customer volume, churn trends, and key contributing factors to customer attrition.

Key Highlights

- **10K total customers** were recorded between 2016 and 2019.
- **2,037 customers exited**, resulting in a **20.4% churn rate**, which remains **above the 15% target** throughout all observed years.
- The **average customer tenure** among both active and churned customers is **4.86 years**, showing moderate retention.

Country-Level Insights

- **Germany** recorded the **highest churn rate** at **32.4%**, followed by **Spain (16.7%)** and **France (16.2%)**.
- Despite slight yearly fluctuations, no country managed to achieve a churn rate below the overall target of 16.5%.

Trend Analysis

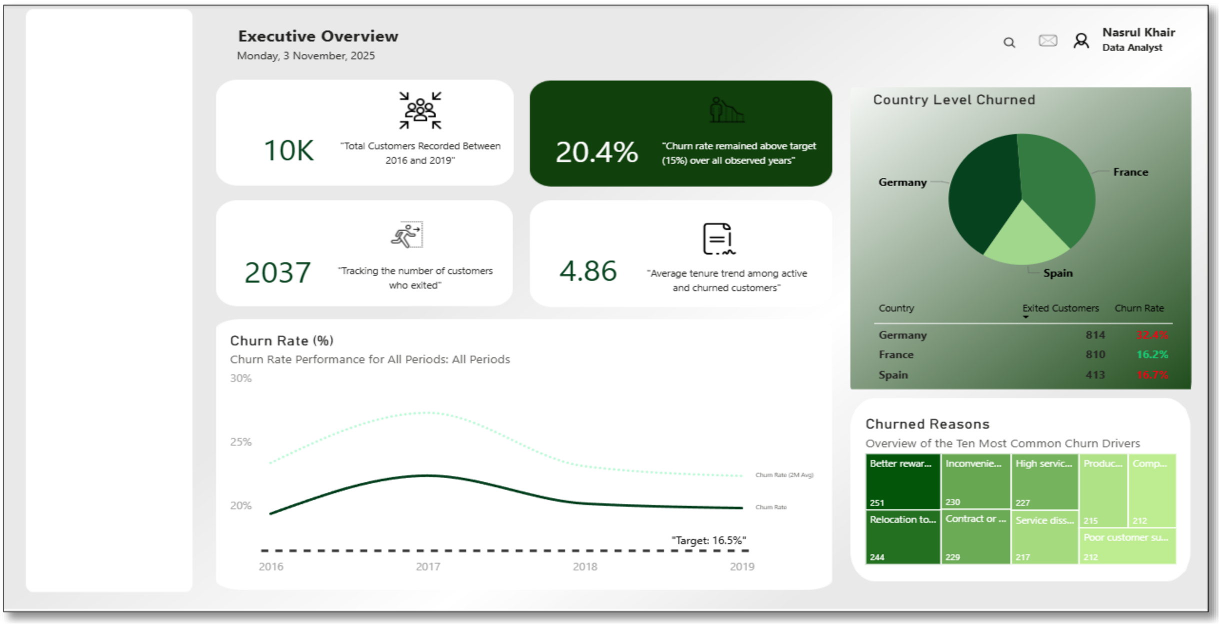
- Churn rate peaked around **2017** before showing gradual improvement towards **2019**, but it remained **above target**.
- The overall churn trend suggests ongoing customer dissatisfaction or weak retention mechanisms despite slight progress.

Churn Drivers

The top churn reasons identified include:

- Better rewards elsewhere (251 cases)
- Inconvenient service process (230)
- High service cost (227)
- Relocation (244)
- Service dissatisfaction (217)
- Poor customer support (212)

These insights highlight that **pricing, service experience, and support quality** are the main contributors to customer churn.



2. Customer Demographic & Profile Analysis

User Story 2: Customer Demographics & Profile Analysis

🎯 Objective

To analyze customer churn patterns through demographic and behavioral segmentation — including age, gender, geography, activity status, and credit card ownership — and uncover which customer groups contribute most to churn.

📊 Page Overview

The **Customer Demographics & Profile Analysis** page provides a deeper understanding of the customer base composition and its relationship with churn.
It complements the Executive Overview by focusing on **who the customers are**, **how they differ**, and **which profiles are most at risk of churn**.

🧠 Key Insights

- **Age-based churn trends:**
  Customers aged **25–45** represent the **largest and most churn-prone segment**, showing higher volatility in churn rates compared to other age groups.

- **Gender analysis:**
  While the gender distribution is balanced, **female customers show a slightly higher churn rate (~21%)**, suggesting potential differences in service expectations or engagement.

- **Geographical credit profiles:**
  German customers tend to have **lower median credit scores** than those in France and Spain, possibly contributing to their **higher churn rate** observed in the Executive Overview.

- **Activity status impact:**
  Inactive members churn **2–3× more** than active ones, confirming that engagement plays a critical role in retention.

- **Credit card ownership:**
  Non-cardholders churn significantly more, highlighting a **cross-sell opportunity** — encouraging customers to adopt credit products could help reduce churn.

- **Tenure distribution:**
  The majority of customers fall within **3–5 years of tenure**, which aligns with a typical retention cycle — intervention at this stage can have a major impact on churn prevention.

🧩 Visual Layout Summary

Section	**Visual**	**Purpose / Expected Insight**
Top Section	KPI Cards: Total Customers, Active %, Credit Card % + Slicer Panel	High-level profile snapshot and filtering options
Middle Section	1️⃣ Age vs Churn Rate (Clustered Column + Line)	Identify age groups most likely to churn
	2️⃣ Gender vs Churn (100% Stacked Bar)	Compare churn share by gender
	3️⃣ Geography vs Credit Score (Box/Violin Plot)	Observe credit score distribution across regions
Bottom Section	4️⃣ Active vs Inactive (Clustered Column)	Measure churn by engagement status
	5️⃣ Credit Card Ownership vs Churn (Bar Chart)	Evaluate churn impact by product ownership
	6️⃣ Tenure Distribution (Histogram)	Assess customer retention pattern by years
Advanced (Optional)	7️⃣ Customer Persona Clustering (Scatter + Clustering)	Identify high-risk churn personas

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🧮 Key DAX Measures

Measure	Formula	Description
Age Group	Age Group = SWITCH(TRUE(), [Age]<26,"18–25",[Age]<36,"26–35",[Age]<46,"36–45",[Age]<56,"46–55","56+")	Categorize customers into age bins
Churn Count	CALCULATE(COUNTROWS(FACT_CUSTOMER_CHURN_V1), DIM_EXIT_CUSTOMER[ExitCategory]="Exit")	Counts churned customers
Churn Rate by Attribute	DIVIDE([Churn Count],[Total Customers])	Context-specific churn rate
Active %	DIVIDE([Active Members],[Total Customers])	Share of active members
Credit Card %	DIVIDE([Credit Card Holders],[Total Customers])	% of customers with credit cards

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🎨 Design & Styling

- Maintain **visual consistency** with Page 1 (Executive Overview).
- Use **rounded white KPI cards** with light shadows for clarity.
- Keep **navigation on left panel** and adopt **light green accents** for segmentation visuals.
- Include small **icons** for visual guidance (🧑 demographics, 💳 credit, 🌍 geography).

📖 Narrative Summary

The demographic analysis reveals that churn is **concentrated among mid-age customers (25–45)** and **inactive, non-credit-card holders**.
Behavioral engagement and product adoption emerge as **key levers for retention**, while credit score and geography offer additional predictive context.
Future initiatives should focus on **activating dormant customers**, **incentivizing credit card usage**, and **personalizing retention campaigns** for high-risk profiles.

Objective

Overview

To analyze customer churn patterns through demographic and behavioral segmentation — including age, gender, geography, activity status, and credit card ownership — and uncover which customer groups contribute most to churn.